

Simulate MOSFET-Based Ideal Diode

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A simple explanation for the working of an ideal diode circuit based on P-channel MOSFET and PNP transistor is presented here. Typical voltage required to turn on a diode is forward voltage. An ideal diode should have zero forward voltage but, unfortunately, there is no such thing as an ideal diode.

A real diode has forward voltage drop and reverse leakage current. Even a Schottky diode has a forward voltage drop between 0.2V and 0.4V, which can be significant, especially when used in series with batteries with lower voltages that reduce the efficiency and life of the batteries.

Many diode applications require reverse current blocking, most common application being battery-operated circuits. Another example of diode application is reverse voltage or reverse current protection for USB supply as used in Raspberry Pi 3.

Limitations of a practical diode call for the need of a near ideal diode with as little forward voltage drop as possible. While selecting a practical diode, its forward voltage and reverse leakage current should be as low as possible. Simulation and testing results in this article showcase the working of the typical circuit of an ideal diode or smart diode. Details for designing the circuit are also provided to enable you to easily adapt the circuit to your designs.

Circuit and working

The circuit consists of P-channel enhancement mode MOSFET (Q1), two PNP transistors (Q2A and Q2B) and three resistors (R1, R2 and R3), as shown in Fig. 1.

Q1 is the series element, and the two bipolar junction transistors, or BJTs (Q2A and Q2B) are the control elements that turn Q1 on or off. When Q1 is on, it acts as a diode in forward bias condition, and when it is off, it acts as a diode in reverse bias condition.

Source of Q1 is connected to cathode side, and the drain to anode side, to ensure the body of MOSFET is reverse-biased. The gate is controlled by cathode side BJT (Q2B). When Q2B is on, Q1 will be off, and vice-versa.

Both BJTs are designed to work as switches. Emitters of BJTs are connected to anode and cathode

sides, and bases are tied together. Base-emitter junctions of the two transistors effectively become OR-ing diodes for voltages at anode and cathode sides. Base-emitter junction of the transistor at the higher voltage side will be forward-biased, while the other will be reverse-biased.

Since BJTs are configured to work as switches, the one with the higher voltage at emitter will be on, thereby forcing the other BJT to be off. This is the simplicity of the circuit. When voltage at cathode side is higher, Q2B will be on and Q1 will be off, making the diode reverse-biased.

Base-emitter voltage drop (V_{BE}) of the transistors needs to be as equal as possible to ensure the circuit reacts to the slightest difference in cathode and anode side voltages. It is highly recommended to use a matched pair PNP transistor for the application.

The circuit in Fig. 1 will take the shape of a current mirror if emitters of both BJTs are shorted. In a current mirror, current through the right-side transistor will depend only on current through the left-side transistor, for a given temperature. However, in this circuit, emitters do not get shorted even when Q1 is on, since Q1 connected in series introduces a voltage drop that depends on the gate to source voltage (V_{GS}) and drain current (I_D) of Q1.

That is, with the introduction of Q1, more parameters get introduced. Voltage drop across transistor Q2B is effectively gate-to-source voltage (V_{GS}) of Q1. To keep Q1 in saturation, voltage drop across Q2B needs to be more than the threshold voltage (V_T) of Q1.

Another parameter that depends on V_{GS} is current through Q1. Thus, Q2B goes to active region instead of cut-off region with a value of collector current balancing the different parameters. Calculating

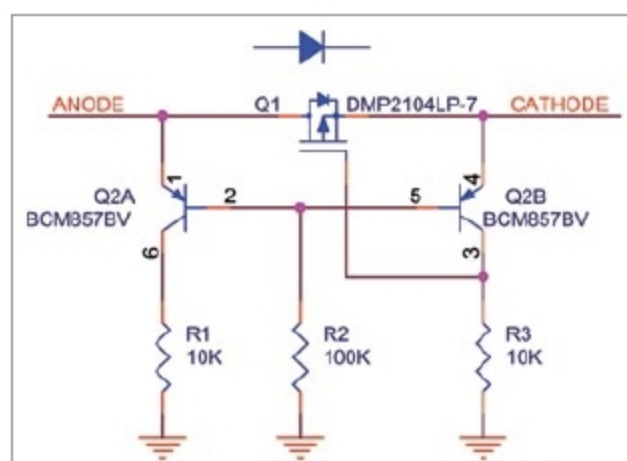


Fig. 1: Ideal diode schematics

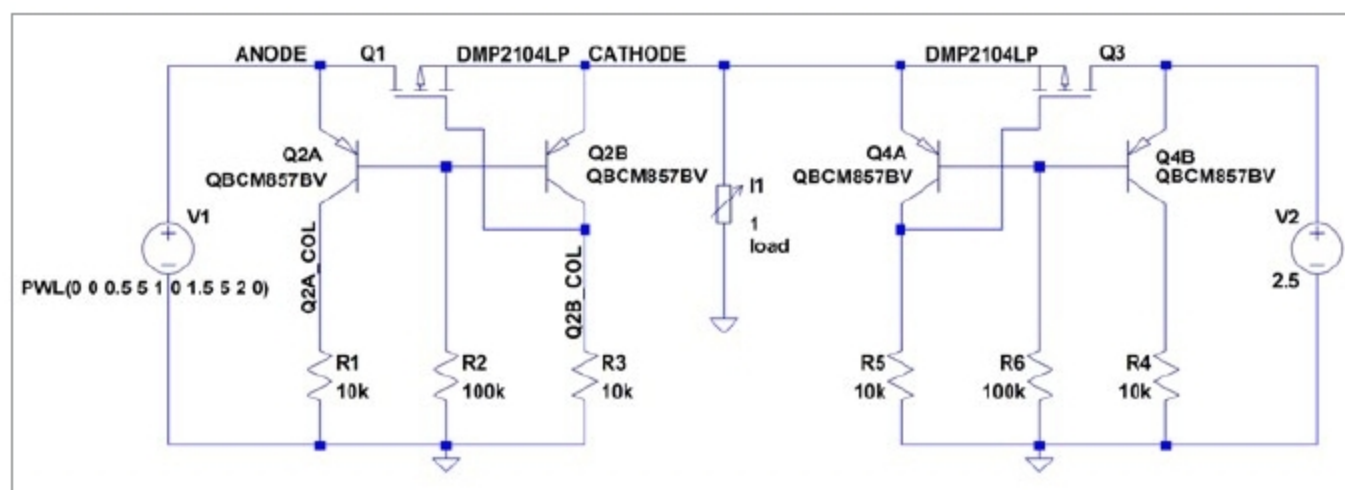


Fig. 2: Ideal diode schematics in LT SPICE environment